

VOLUME 4 OF 4 · MODULE REFERENCE

SADHANA

Module build guide: eight modules, each start to finish in one chapter

Per module · Data flow, file tree, prompt sequence

Per task · Copy-paste prompt, verify, done-when

Per gate · Test checklist before you move on

ఒక్కో మాడ్యూల్ మొదటి నుంచి చివరి వరకు

ఒకే చాప్టర్లో

Contents

Vol 3 vs Vol 4 — which one to open	4
Read this before you start	4
The context primer	5
Module index	6
M0 — FOUNDATION	6
M0 data flow	6
M0 file tree	7
M0 build sequence	8
M0 test checklist	13
M0 exit gate	14
M1 — EXAM HUB	15
M1 data flow	15
M1 file tree	16
M1 build sequence	16
M1 test checklist	19
M1 exit gate	20
M2 — NOTIFICATIONS AND ELIGIBILITY	21
M2 data flow	21
M2 file tree	22
M2 build sequence	23
M2 test checklist	25
M2 exit gate	26
M3 — INGESTION	27
M3 data flow	27
M3 file tree	28
M3 build sequence	28
M3 test checklist	30
M3 exit gate	30
M4 — QUIZ AND STREAKS	31
M4 data flow	31

M4 file tree	32
M4 build sequence	33
M4 test checklist	35
M4 exit gate	36
M5 — PUSH AND WHATSAPP	37
M5 data flow	37
M5 file tree	38
M5 build sequence	38
M5 test checklist	40
M5 exit gate	40
M6 — PWA AND PERFORMANCE	41
M6 data flow	41
M6 file tree	42
M6 build sequence	42
M6 test checklist	43
M6 exit gate	43
M7 — COMPLIANCE AND LAUNCH	44
M7 data flow	44
M7 file tree	45
M7 build sequence	45
M7 test checklist	46
M7 exit gate	47
Reusable prompt library	48
The five decisions you will be tempted to reverse	50

Field	Detail
Document	Volume 4 of 4 — Module-wise Build Guide
Companions	Vol 1 R&D · Vol 2 Implementation · Vol 3 Sprint Playbook
Version	1.0
Date	September 2026
Owner	Lokesh
Covers	8 modules, 62 tasks, every file, every prompt

Vol 3 vs Vol 4 — which one to open

Same 62 tasks. Different cut.

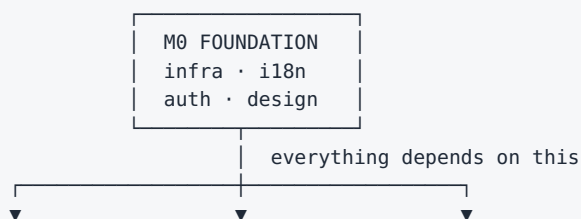
	Vol 3 — Sprint Playbook	Vol 4 — this document
Organised by	Time (Week 1 → Week 14)	Module (M0 → M7)
Answers	"What do I do this Monday?"	"How does the quiz module work, whole thing?"
Use when	Planning your week, tracking the gate	Building or debugging one module
Extra content	Sprint exit gates, weekly ritual	Data flow, file tree, test checklist per module

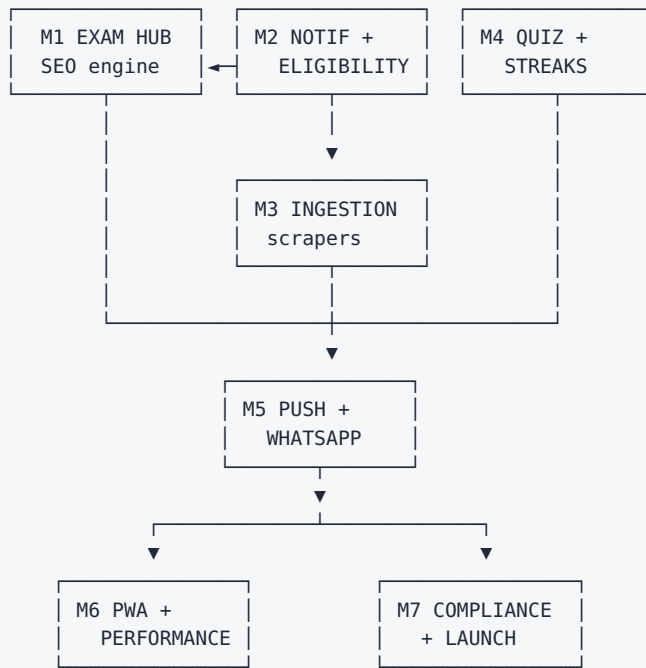
Keep Vol 3 open for scheduling. Keep this one open while coding.

Read this before you start

సారాంశం: మాడ్యూల్ వరుసలో కట్టు, కానీ ఆధారపడే క్రమాన్ని పాటించు.

Module order is not free order. M0 must be complete before anything else compiles. After that the dependency graph below governs.





Recommended build order: M0 → M1 → M2 → M3 → M4 → M5 → M6 → M7.

M2 needs M1's exam records to attach notifications to. M3 needs M2's notification model to write into. M5 needs M2 and M4 to have something worth pushing about.

The context primer

Paste at the start of every Claude session. Without it you get generic Laravel code that does not fit your schema.

PROJECT CONTEXT – SADHANA

Product: Telugu-first portal for Indian government/private job aspirants.
 Modules: notification feed with eligibility matching, exam hub pages (SEO), daily quiz with streaks, material library, doubt community, test series.

Stack:

- Laravel 12, PHP 8.3
- Livewire 3 + Alpine.js (server-rendered – SEO is the business model, so no SPA, no client-side routing)
- Filament 3 for admin
- MySQL 8, Redis (cache/queue/streaks), Meilisearch (per-locale indexes)
- Cloudflare R2 for files, Cloudflare CDN
- Tailwind CSS, Pest for tests

Hard rules:

1. Multi-language (te/en) is core. Translatable content uses spatie/laravel-translatable JSON columns. Slugs are NEVER translated.
2. Every cache key must include app()->getLocale().
3. Target device is a Rs 8,000 Android on 3G. JS budget under 100 KB. Telugu text runs 15-30% longer than English.
4. Modular monolith. Modules talk via service classes, never by reaching into each other's models.
5. No pirated content anywhere.

Code style: Laravel conventions, strict types, service classes for logic, form requests for validation, Pest for tests. Readable over clever.

Today I am working on: [FILL IN]

Module index

#	Module	Tasks	Days	Owns
M0	Foundation	T01-T12	10	Infra, i18n, auth, design system, CI/CD
M1	Exam Hub	T13-T22	10	Exam pages, SEO, search, caching
M2	Notifications + Eligibility	T23-T31	12	Feed, eligibility engine, saves, reminders
M3	Ingestion	T32-T39	6	Scrapers, AI extraction, review queue
M4	Quiz + Streaks	T40-T50	12	Question bank, daily quiz, streaks, leaderboards
M5	Push + WhatsApp	T51-T54	5	Web push, WhatsApp Cloud API, dispatch
M6	PWA + Performance	T55-T57	5	Service worker, offline, load testing
M7	Compliance + Launch	T58-T62	8	DPDP, legal, analytics, beta, launch

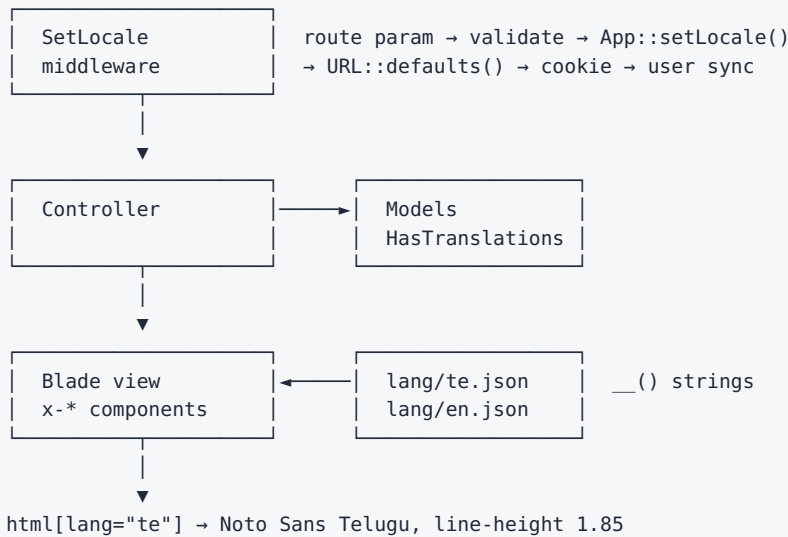
M0 — FOUNDATION

సారాంశం: పునాది. ఇక్కడ తొందరపడితే మిగతా 12 వారాలు బాధపడాలి.

Purpose	Everything else compiles on top of this
Tasks	T01-T12 · 10 working days
Depends on	Nothing
Unlocks	All modules
Exit signal	An admin creates a bilingual exam in Filament and it renders correctly in Telugu on a real phone

M0 data flow

Request /te/exams
|
▼

**ADMIN SIDE**

/admin (Filament)
↓
Tabs per locale
from config/locales.php
↓
publish confirmation
modal

AUTH SIDE

phone → OTP → verify
↓
user + empty profile
↓
profile setup (3 steps)
↓
fields the eligibility
engine will read

M0 file tree

config/locales.php	← supported languages, active flags
app/Http/Middleware/SetLocale.php	
app/Support/Locale.php	← cacheKey() helper
routes/web.php	← locale-prefixed group
lang/te.json	
lang/en.json	← ~120 UI strings
database/migrations/	
app/Models/	← 22 tables
	← User, Profile, Exam, ExamNotification, Question, DailyQuiz, Streak, ...
database/factories/	
database/seeder/	
app/Livewire/Auth/PhoneLogin.php	
app/Livewire/Auth/VerifyOtp.php	
app/Livewire/Profile/SetupWizard.php	
app/Services/Otp/OtpChannel.php	← interface
app/Services/Otp/LogDriver.php	
app/Services/Otp/WhatsAppDriver.php	← stub until M5
app/Filament/Resources/ExamResource.php	
app/Filament/Resources/ExamNotificationResource.php	
resources/views/components/	← x-button, x-card, x-badge, x-input, x-empty-state, x-skeleton

```
resources/views/layouts/app.blade.php
resources/css/app.css ← @font-face, html[lang] rules
tailwind.config.js
public/fonts/ noto-sans-telugu-subset.woff2

app/Http/Controllers/HealthCheckController.php
app/Console/Commands/HealthCheck.php
.github/workflows/ci.yml
.github/workflows/deploy.yml
```

M0 build sequence

T01 — Competitor teardown (*no code*)

Real budget Android, not an emulator. Install Adda247, Testbook, Sakshi Education. Full signup on each. Switch to Telugu, screenshot every silent English fallback. Follow one live notification across all three plus two Telegram channels, time each.

Every moment you feel confused is a bug you must not ship. That list becomes your UX spec.

```
[CONTEXT PRIMER]
Today I am working on: turning competitor research into a spec.

I did a teardown of three Indian exam-prep apps. My raw notes:

[paste notes]

Turn this into:
1. A prioritised list of UX failures I should NOT repeat
2. Three things they do well that I should copy
3. The single biggest gap I can exploit

Specific and concrete. No generic UX advice.
```

Done when: 30+ screenshots and 20+ friction notes written down.

T02 — Project scaffold

```
[CONTEXT PRIMER]
Today I am working on: initial project scaffold.

Exact command sequence for a fresh Laravel 12 project with:
Livewire 3, Filament 3 (admin on /admin, separate guard), Tailwind + Vite,
Pest 3 replacing PHPUnit, Pint, PHPStan level 6, and these packages:
spatie/laravel-translatable, laravel/scout, meilisearch-php,
spatie/laravel-sitemap, intervention/image, sentry/sentry-laravel,
propaganistas/laravel-phone

Then give me:
1. Final composer.json
2. phpstan.neon
3. pint.json
```

4. `.env.example` with every variable this project needs

Commands only, no explanation of what Laravel is.

Verify: `php artisan serve` loads · `/admin` shows Filament login · `php artisan test` passes.

T03 — Migrations (*run as 3 separate sessions*)

One session producing 22 migrations will drift — naming goes inconsistent by the last few tables.

Batch	Tables
1	users, profiles, user_exam_preferences, exam_categories, exams, exam_cutoffs
2	notifications, notification_saves, questions, daily_quizzes, quiz_attempts, streaks
3	materials, posts, answers, votes, content_translations, ad_campaigns, ad_events, scrape_sources, translation_queue, push_tokens

```
[CONTEXT PRIMER]
Today I am working on: database migrations, batch [N] of 3.

Target schema:
[paste the SQL for this batch from Vol 2 Chapter 4]

Convert to Laravel 12 migrations:
- One migration per table, ordered correctly for foreign keys
- $table->json() for translatable columns
- Keep every index I specified – they are deliberate
- softDeletes where I marked deleted_at
- foreignId()->constrained() where it reads cleanly

Full file contents and the filename for each.
```

Verify: `php artisan migrate:fresh` clean · `SHOW INDEX FROM notifications;` shows your indexes · rollback works.

T04 — Models, relationships, factories

```
[CONTEXT PRIMER]
Today I am working on: Eloquent models.

My migrations:
[paste 4-5 related migrations]

Generate matching models with:
- HasTranslations trait + $translatable array
- All relationships, both directions
- $casts for dates, json, booleans
- SoftDeletes where present
- Scopes: scopePublished, scopeActive, scopeForLocale
```

- A factory each with REALISTIC Indian exam data – real exam names, real Telugu strings, realistic vacancy counts and dates

Full files with paths.

Verify:

```
php artisan tinker
>>> Exam::factory()->count(5)->create()
>>> Exam::first()->notifications
```

T05 — Locale routing

[CONTEXT PRIMER]

Today I am working on: multi-language routing.

Build:

1. config/locales.php – locales with name, native name, active flag. te and en active; hi and ta defined but inactive.
2. SetLocale middleware:
 - read {locale} route param, validate against active, fall back
 - App::setLocale()
 - URL::defaults(['locale' => \$locale]) so route() never needs it passed
 - 1-year cookie
 - sync user's preferred_locale without firing events
3. routes/web.php: locale-prefixed group, where clause, named routes for home, exams.index, exams.show, notifications.index, notifications.show, quiz.today
4. Root "/" resolution: cookie → Accept-Language → 'te'
5. Language switcher component linking to the SAME page in the other locale, not the homepage

Full files plus Pest tests for the middleware.

Verify: switching language on an exam detail page stays on that exam.

T06 — UI translation files

[CONTEXT PRIMER]

Today I am working on: UI string translations.

Generate lang/te.json for a government-job portal. Use ENGLISH SENTENCES as keys (not dot notation) so missing keys fall back readably.

~120 strings covering: navigation, auth/OTP, profile form, notification feed, eligibility badges, exam hub sections, quiz UI, streaks, errors, empty states, buttons, dates.

Critical rules:

- "Notification" stays "నోటిఫికేషన్", NOT "ప్రకటన" – that is what aspirants actually search for
- "Group 2" stays "గ్రూప్ 2"
- Exam names and abbreviations (SSC, RRB, TGPSC) untranslated

- Everyday register, not formal literary Telugu
- Support Laravel :placeholder syntax

Full JSON. Also lang/en.json with identical keys mapping to themselves.

Done when: a Telugu speaker confirms it reads naturally, not machine-translated.

T07 — Design system and Telugu typography

[CONTEXT PRIMER]

Today I am working on: design system.

Tailwind + Blade design system for a Telugu exam portal.

Constraints:

- Rs 8,000 Android, 360px viewport, 3G
- Telugu strings run 15-30% longer and glyphs are taller – every component must survive that without clipping
- Self-hosted subset Noto Sans Telugu, NOT Google Fonts CDN
- Numerals stay Latin (2026, Rs 28,940)

Give me:

1. tailwind.config.js – colours (trustworthy, not playful; this is a career decision tool), type scale, spacing
2. @font-face plus html[lang="te"] rules, line-height 1.85 Telugu / 1.6 English
3. Blade components: x-button (3 variants), x-card, x-badge (green/amber/grey for eligibility), x-input, x-select, x-empty-state, x-skeleton
4. Base layout: header, mobile-first bottom nav, language switcher

Render EVERY component with both English and Telugu text so I can see the layout hold.

Verify: open every component at 360px with Telugu content. Nothing clips.

T08 — Phone OTP auth

[CONTEXT PRIMER]

Today I am working on: phone OTP auth.

Phone-number authentication for Indian users:

1. Livewire: enter phone → OTP → verify → logged in
2. Indian phone validation (propagandistas/laravel-phone, IN region)
3. 6-digit OTP, 10-min expiry, stored HASHED in cache not the database
4. Rate limits: 3 OTP requests per phone per hour, 5 verify attempts per OTP
5. OtpChannel interface with LogDriver (dev) and WhatsAppDriver (prod, Meta Cloud API) – I wire WhatsApp in M5
6. New user → user + empty profile → profile setup
Returning → intended
7. All copy through __()

No passwords anywhere. Full files plus Pest tests for rate limits and expiry.

Verify: spam the OTP endpoint — rate limiting must actually block.

T09 — Profile setup

These fields exist **only** because M2's eligibility engine reads them.

[CONTEXT PRIMER]

Today I am working on: user profile setup.

Multi-step profile form. Fields (they exist only because the eligibility engine needs them): date_of_birth, gender, category (general/obc/sc/st/ews), is_pwd, is_ex_serviceman, highest_qualification, qualification_stream, state, district

Requirements:

- 3 steps: basic → qualification → exam interests
- Progress indicator; each step saves so nothing is lost on a dropped connection
- District dropdown from selected state (TS and AP districts)
- Exam interests: searchable multi-select from the exams table
- Skippable, but a persistent banner explaining eligibility matching needs it
- DPDP consent: itemised checkboxes with purpose stated; store consent version and timestamp
- 360px first, all copy through __()

Full files, Form Request validation, Pest tests.

Verify: drop the connection mid-wizard — the earlier step must survive.

T10 — Filament resources

[CONTEXT PRIMER]

Today I am working on: Filament admin.

Resources for Exam and ExamNotification.

KEY REQUIREMENT — translatable fields need a TAB PER ACTIVE LOCALE, generated from config('locales.supported'), not hardcoded. English required, Telugu optional but visibly flagged when missing.

ExamResource: name, slug (auto from English name, NEVER translated), category, conducting body, official website, description, exam pattern, syllabus, meta title/description, FAQ repeater.

ExamNotificationResource: all notification fields, grouped into sections: Basic, Vacancy, Eligibility, Dates, Fees, Links, Status. Eligibility needs a key-value field for age_relaxation by category.

Both need:

- Status column with colour coding
- A "missing Telugu translation" filter — content ops need this daily
- Bulk publish action
- A REQUIRED confirmation modal on publish:

"Have you verified dates and eligibility against the official PDF?"

Full resource files.

Done when: the publish confirmation cannot be bypassed.

T11 — CI/CD

[CONTEXT PRIMER]

Today I am working on: CI/CD.

Two GitHub Actions workflows for Laravel 12 deploying to a Hostinger VPS over SSH.

ci.yml (every push and PR): PHP 8.3, MySQL 8 + Redis services, composer install, migrate, pint --test, phpstan, pest --parallel, npm build, and a step that FAILS the build if gzipped JS exceeds 100 KB.

deploy.yml (push to main): maintenance mode, pull, composer install --no-dev, migrate --force, npm build, config/route/view cache, queue restart, supervisor restart, up. Must roll back and exit non-zero if migration fails.

Also the supervisor config for queue workers with priority queues: high, push, default, low.

Done when: a failing test blocks deployment.

T12 — Monitoring baseline

[CONTEXT PRIMER]

Today I am working on: monitoring.

For a Laravel 12 production app:

1. Sentry with release tracking and user context (user id ONLY – never phone or date of birth, DPDP)
2. Laravel Pulse
3. HealthCheckController at /up returning JSON: database, redis, meilisearch, queue depth, last successful scrape time
4. An artisan command sadhana:health alerting when error rate > 1% for 5 min, p95 > 2s, queue depth > 10000, any scraper failed 3 runs
5. Scheduler entries for all of it

Full files.

Verify: stop Redis — `/up` must report it honestly.

M0 test checklist

- `❑ migrate:fresh --seed` produces a working bilingual dataset

- ☐ SetLocale falls back correctly on an unsupported locale
- ☐ `route('exams.show', ['slug' => $s])` works without passing locale
- ☐ Language switcher preserves route parameters
- ☐ No raw translation keys visible on `/te/`
- ☐ Every component holds with a Telugu string 30% longer than English
- ☐ OTP rate limits block; OTPs expire
- ☐ Profile wizard survives a dropped connection
- ☐ Publish confirmation cannot be skipped
- ☐ Failing test blocks deploy
- ☐ `/up` reports honestly when a dependency is down

M0 exit gate

Do not start M1 until all of these are true. Every later module assumes them.

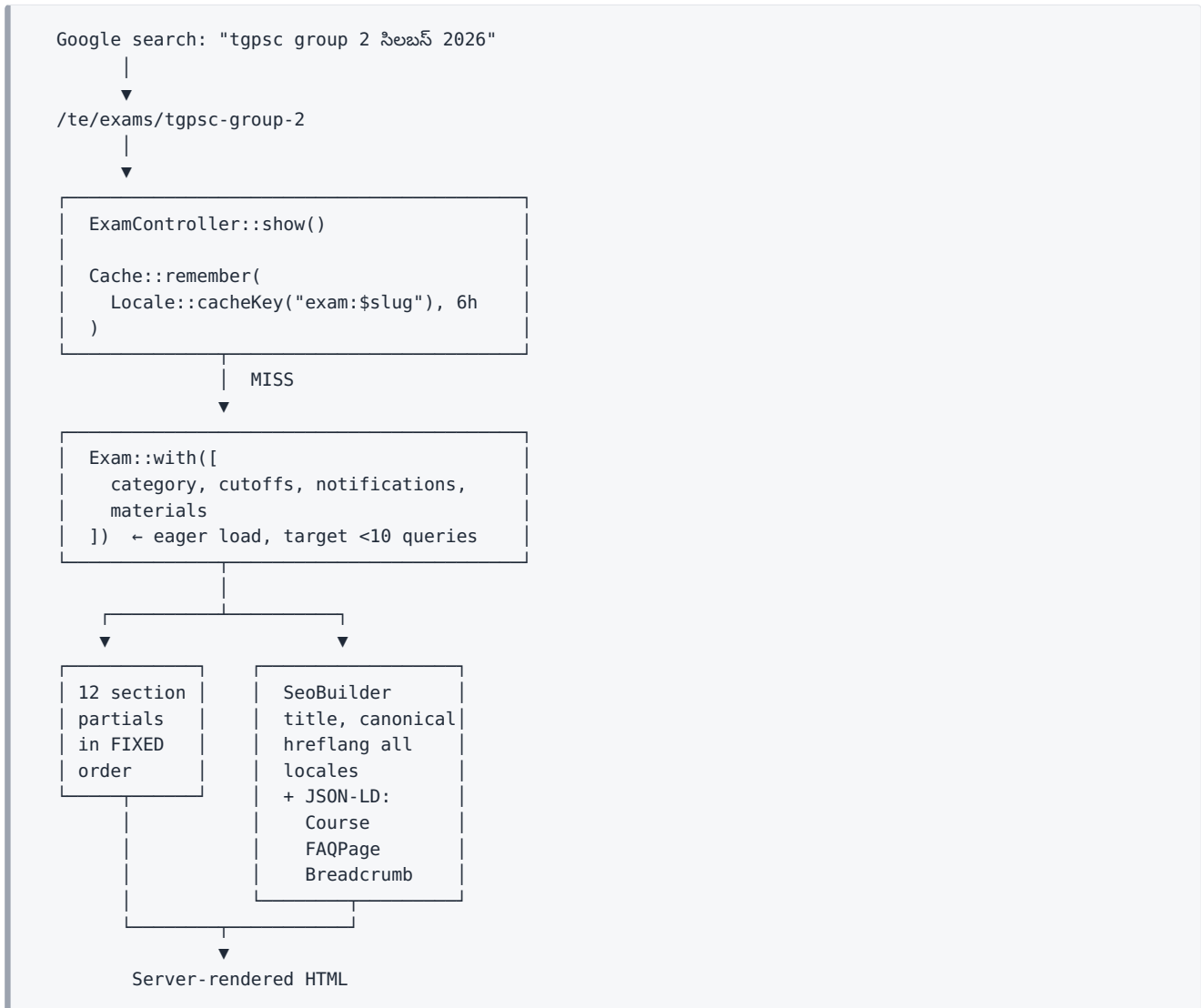
- ☐ `/te/` and `/en/` both render
 - ☐ Admin creates an exam in Filament → it appears on the site in Telugu
 - ☐ Phone signup works end to end
 - ☐ Push to main deploys automatically
 - ☐ A page renders correctly on a real budget Android
-

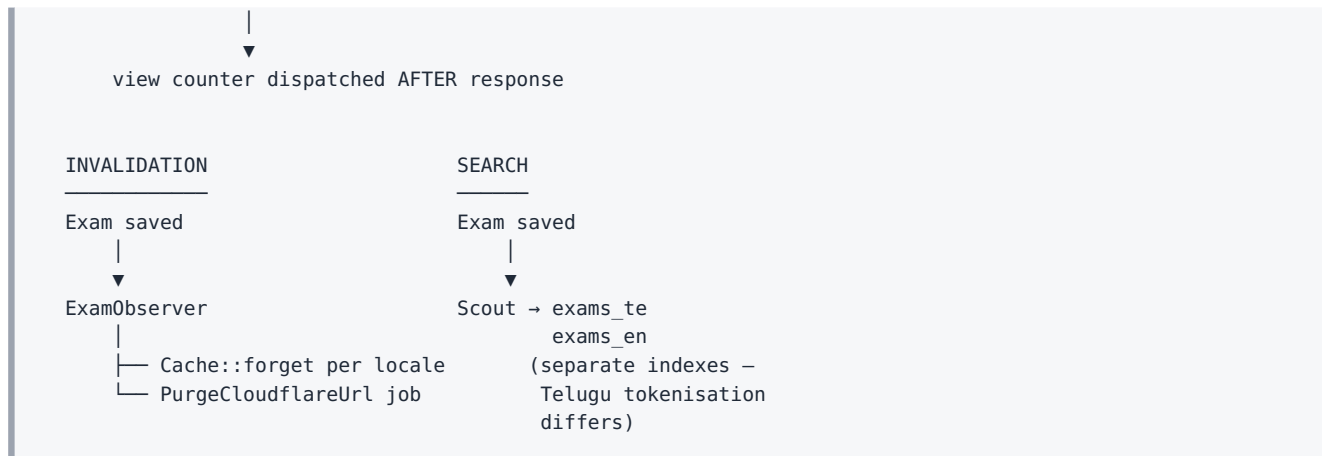
M1 — EXAM HUB

సారాంశం: ఈ పేజీలో గూగుల్ నుంచి ఉచిత ట్రాఫిక్ తెస్తాయి. 60% యూజర్లు ఇక్కడి నుంచే.

Purpose	The SEO engine — 60% of Year-1 acquisition
Tasks	T13-T22 · 10 working days
Depends on	M0
Unlocks	M2 (notifications attach to exams)
Exit signal	10 exam pages live, indexable, structured data validating

M1 data flow





M1 file tree

```

app/Http/Controllers/ExamController.php
app/Livewire/Exam/ExamFilters.php
app/Support/SeoBuilder.php
app/Support/StructuredData.php
app/Observers/ExamObserver.php
app/Jobs/PurgeCloudflareUrl.php
app/Console/Commands/GenerateSitemaps.php
app/Console/Commands/ReindexAllLocales.php
app/Livewire/Search/GlobalSearch.php

resources/views/exam/index.blade.php
resources/views/exam/show.blade.php
resources/views/exam/partials/
  latest-notification.blade.php
  quick-facts.blade.php
  eligibility.blade.php
  exam-pattern.blade.php
  syllabus.blade.php
  previous-papers.blade.php
  cutoffs.blade.php
  free-material.blade.php
  community-questions.blade.php ← placeholder until Phase 2
  related-exams.blade.php
  faq.blade.php
resources/views/layouts/partials/seo.blade.php

public/robots.txt
  
```

M1 build sequence

T13 — Exam listing

```

[CONTEXT PRIMER]
Today I am working on: exam listing page.

Build /[locale]/exams:
- Grouped by exam_categories with icons
- Card: name, conducting body, live-notification count badge, next date
- Search box filtering CLIENT-SIDE over already-loaded exams – no round
  
```

```

    trip, assume 3G
- Filters: state (TS/AP/National), qualification level
- Server-rendered for SEO; Livewire only for filters
- Cached 6h with locale in the key

Controller, Livewire component, Blade view, and a Pest test asserting a
Telugu exam name renders on the /te/ route.

```

Verify: view page source — exam names must be in the HTML. Disable JS; the list must still show.

T14 — Exam detail page

```

[CONTEXT PRIMER]
Today I am working on: exam detail page – my main SEO asset.

/[locale]/exams/{slug} with sections IN THIS EXACT ORDER:
1. H1: exam name + current year
2. Latest notification status banner
3. Quick facts table (body, vacancies, dates, qualification)
4. Eligibility
5. Exam pattern table
6. Syllabus – expandable per subject, PDF download
7. Previous year papers
8. Cutoff trends by year and category
9. Recommended free material
10. Top community questions (placeholder – Phase 2)
11. Related exams
12. FAQ accordion

Requirements:
- Fully server-rendered. Alpine only for accordions.
- Eager load everything – ONE page, not an N+1 disaster
- Cache 6h, locale in key
- View counter incremented AFTER the response, never blocking it
- Semantic HTML: article, section, correct heading hierarchy

Controller, one Blade partial per section, Pest test.

```

Verify: Debugbar query count under 10 · page under 500ms uncached.

T15 — SEO infrastructure

```

[CONTEXT PRIMER]
Today I am working on: SEO. This is my primary acquisition channel –
get it exactly right.

Build:
1. App\Support\SeoBuilder with forExam(), forNotification(), forHome().
   Each returns title, description, canonical, alternates keyed by locale.
2. A seo Blade partial: title, meta description, canonical, hreflang for
   every active locale plus x-default → English, Open Graph.
3. Structured data builders:
   - Course for exams
   - JobPosting for notifications (gets me into Google Jobs – include
     validThrough, hiringOrganization, jobLocation, baseSalary,

```

```

    totalJobOpenings)
    - FAQPage for exam FAQ blocks
    - BreadcrumbList site-wide
4. Sitemaps: separate per locale per content type, under a sitemap index,
   nightly artisan command.
5. robots.txt allowing everything except /admin and /api.

Full files. Show me the rendered head for one Telugu exam page.

```

Verify: Google Rich Results Test — JobPosting and FAQPage must both validate with zero errors.

T16 — Cache and purge

```

[CONTEXT PRIMER]
Today I am working on: caching.

Build:
1. App\Support\Locale::cacheKey($key) appending the current locale.
   Every cache call in this project goes through it.
2. Observers on Exam and ExamNotification that on save/delete forget the
   local cache for EVERY active locale and dispatch a Cloudflare purge
   for the matching URLs.
3. PurgeCloudflareUrl queued job with retry and failure logging.
4. The exact Cloudflare page rules I should configure: exam pages 6h,
   notification detail 1h, anonymous feed 5min, never cache /admin or
   authenticated sessions.

Full files plus a Pest test proving a Telugu cache entry clears on save.

```

Verify: edit an exam, hard-refresh `/te/` and `/en/` — both must update.

T17-T20 — Content population (*the real work*)

Ten exams, complete, both languages: TGPSC Group 2, TGPSC Group 4, APPSC Group 2, SSC CGL, SSC CHSL, RRB NTPC, RRB Group D, TS Police Constable, AP Police Constable, DSC/TET.

Per exam: description, eligibility, pattern, full syllabus, 5 years of cutoffs, 3 previous papers, 8 FAQs.

```

[CONTEXT PRIMER]
Today I am working on: content for [EXAM NAME].

I am pasting the official notification PDF text. Extract and structure:
1. A 2-sentence description
2. Eligibility (qualification, age with category-wise relaxation)
3. Exam pattern table (papers, marks, duration, negative marking)
4. Full syllabus by subject and topic
5. 8 FAQs aspirants actually ask, with answers

Rules:
- Use ONLY what is in the text below. If not stated, write "Not specified
  in notification" – do NOT fill gaps from memory.
- Flag anything ambiguous for my manual check.
- English first, then Telugu, keeping exam names, "Group 2" and

```

abbreviations untranslated.

[PASTE PDF TEXT]

Verify: you personally cross-check every date and eligibility line against the official PDF. Twice.

T21 — Search

[CONTEXT PRIMER]

Today I am working on: Meilisearch.

Wire Scout to Meilisearch with SEPARATE INDEXES PER LOCALE (notifications_te, notifications_en, exams_te, exams_en). Telugu tokenisation differs from English; a shared index wrecks relevance for both.

Give me:

1. Searchable config on Exam and ExamNotification: searchableAs() returning the locale-suffixed index, toSearchableArray() pulling the right translation
2. An artisan command reindexing ALL locales, not just the current one
3. Per-locale index settings: searchableAttributes, filterableAttributes, sortableAttributes, ranking rules with recency weighting
4. A global search Livewire component: debounced, grouped by type, keyboard navigable, works at 360px

Full files.

Verify: a Telugu term on `/te/` and an English term on `/en/` both return sensible, different results.

T22 — Search Console (*no code*)

Verify domain · submit each locale's sitemap separately · set international targeting · check the hreflang report.

Done when: both sitemaps submitted, hreflang reports zero errors.

M1 test checklist

- ☐ Exam list renders with JS disabled
- ☐ Detail page under 10 queries
- ☐ Detail page under 500ms uncached
- ☐ hreflang present for every active locale + x-default
- ☐ Course, JobPosting, FAQPage all validate
- ☐ Cache clears for all locales on edit
- ☐ Cloudflare purge fires and retries on failure

- ☐ Sitemaps generate nightly and split by locale
- ☐ Search returns locale-appropriate results
- ☐ Telugu exam name appears in raw HTML source

M1 exit gate

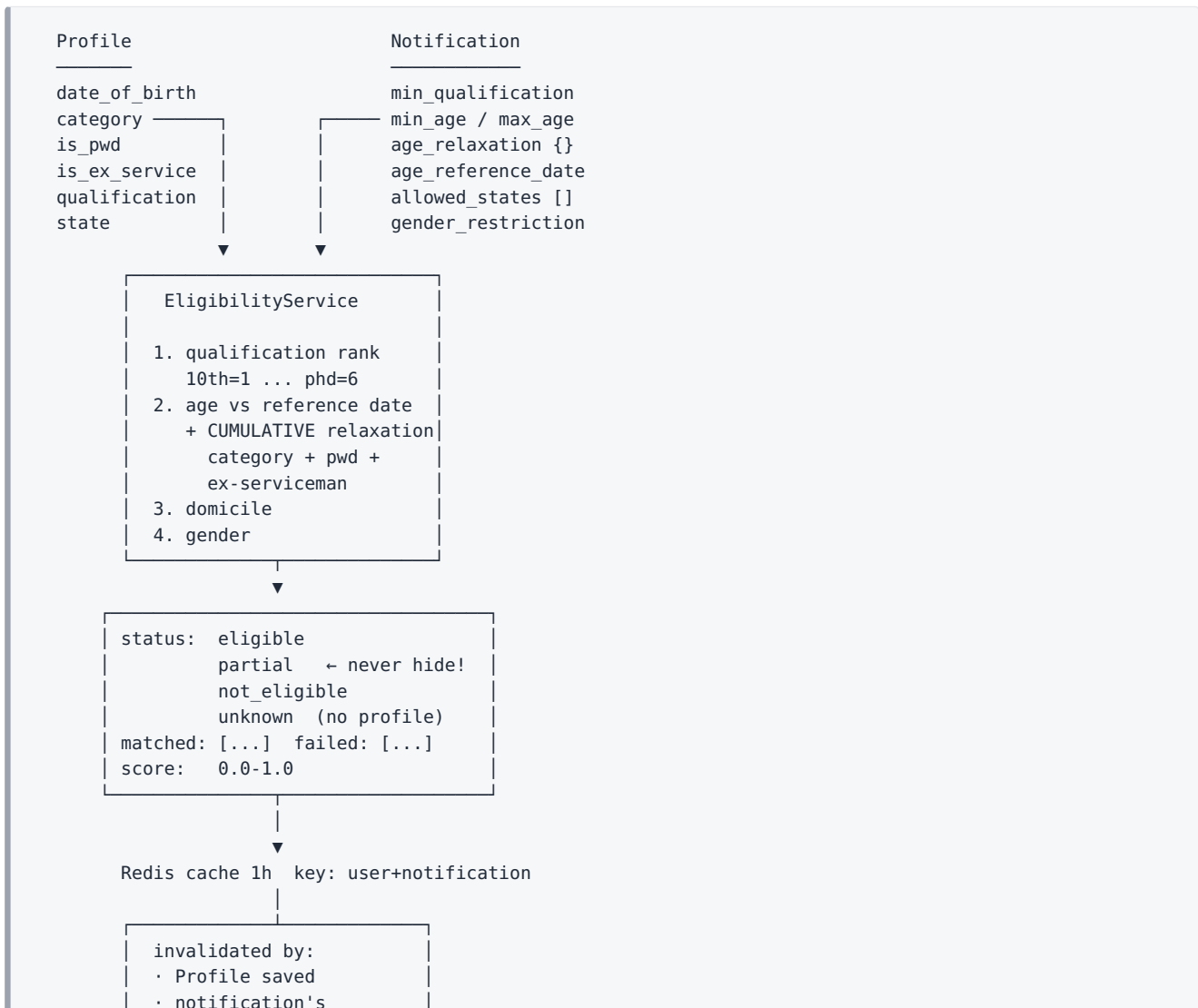
- ☐ 10 exam pages complete in both languages
 - ☐ Rich Results Test passes on all three types
 - ☐ Sitemaps submitted, hreflang clean in Search Console
 - ☐ Every page carries "Last verified" + official source link
-

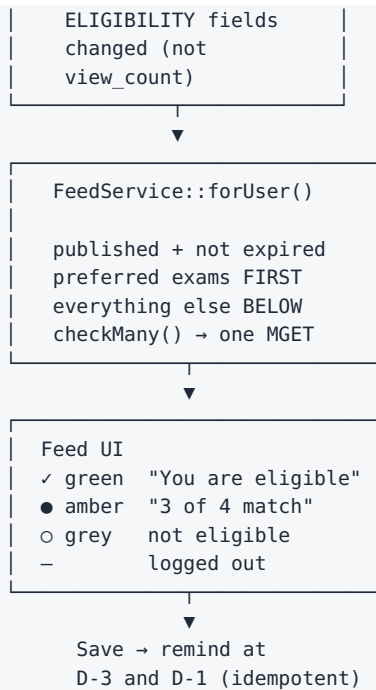
M2 — NOTIFICATIONS AND ELIGIBILITY

సారాంశం: ఉత్పత్తి యొక్క గుండె. అర్హత ఇంజిన్ మన అసలు ప్రత్యేకత.

Purpose	The core value proposition — "should I apply to this?"
Tasks	T23–T31 · 12 working days
Depends on	M0, M1
Unlocks	M3, M5
Exit signal	A user sets a profile and sees a personalised, eligibility-tagged feed

M2 data flow





M2 file tree

```

app/Services/EligibilityService.php      ← highest-value code in the product
app/Services/EligibilityCache.php
app/Services/FeedService.php
app/Http/Controllers/NotificationController.php
app/Livewire/Feed/NotificationFeed.php
app/Livewire/Feed/FilterBar.php
app/Livewire/Notification/SaveButton.php
app/Livewire/Notification/ReportError.php
app/Observers/ProfileObserver.php
app/Jobs/SendDeadlineReminders.php
app/Console/Commands/ExpireNotifications.php

resources/views/feed/index.blade.php
resources/views/feed/partials/card.blade.php
resources/views/feed/partials/filter-bar.blade.php
resources/views/notification/show.blade.php
resources/views/notification/partials/
  eligibility-panel.blade.php
  vacancy-table.blade.php
  dates-timeline.blade.php
  fee-table.blade.php
  how-to-apply.blade.php
resources/views/saved/index.blade.php

tests/Unit/EligibilityServiceTest.php    ← 95%+ coverage required
tests/Feature/FeedTest.php
  
```

M2 build sequence

T23 — Eligibility engine (***budget a full day; test it hardest***)

Ask for **tests first**. It forces the edge cases into the open before the logic is written.

```
[CONTEXT PRIMER]
Today I am working on: the eligibility engine. This is the single
highest-value piece of logic in my product – a wrong answer misleads
someone about their career, so correctness matters more than elegance.

Build App\Services\EligibilityService with
check(ExamNotification, ?Profile) returning:
['status' => 'eligible'|'partial'|'not_eligible'|'unknown',
 'matched' => [...], 'failed' => [...], 'score' => float]

Rules:
1. QUALIFICATION – rank: 10th=1, 12th=2, iti=2, diploma=3, degree=4,
   btech=4, mbbs=5, pg=5, phd=6. User rank >= required.
2. AGE – calculate against age_reference_date, falling back to
   apply_end_date then today. Relaxation from age_relaxation JSON:
   category PLUS pwd PLUS ex_serviceman, CUMULATIVE.
3. DOMICILE – if allowed_states set, user's state must be in it.
   Null = all-India.
4. GENDER – if gender_restriction is not 'any', must match.
5. Return 'partial' when some criteria pass – NEVER hide a notification.
   Users may have relaxations I do not know about.
6. Return 'unknown' with no profile.

Write EXHAUSTIVE Pest tests FIRST, covering:
- SC candidate aged 33, max-age 30, sc:5 relaxation → eligible
- PwD + OBC stacking both relaxations
- Birthday exactly on the reference date
- Notification with no age limits
- Notification with no qualification specified
- Profile with null date_of_birth
- All-India notification vs a TS-only user
- Leap-year birth date

95%+ coverage on this class. Tests first, then the implementation.
```

Verify: `php artisan test --filter=Eligibility --coverage`

T24 — Eligibility caching

```
[CONTEXT PRIMER]
Today I am working on: eligibility caching.

Eligibility runs for every notification on every feed load. Make it cheap:
1. Cache per user+notification for 1h in Redis
2. Invalidate all entries for a user when their profile changes
   (Profile observer)
3. Invalidate all entries for a notification when its ELIGIBILITY fields
   change – but NOT on unrelated changes like view_count
```

4. checkMany(Collection, ?Profile) doing ONE Redis MGET, not N round trips

Full files plus a test proving a profile update clears the cache.

Verify: load the feed twice; the second load must compute zero eligibility checks.

T25 — Feed service

[CONTEXT PRIMER]

Today I am working on: the notification feed query.

App\Services\FeedService::forUser(?User, array \$filters, int \$perPage).

- Published only, not past apply_end_date (null = still open)
- Preferred exams float to the TOP; everything else still visible below
 - never hide content
- Filters: job_type, qualification, district (JSON column match or null), closing_soon (7 days), exam_id
- Eager load exam
- Order: preferred first, then published_at desc
- Paginated

Plus a companion anonymous cached feed (5-min cache, locale-keyed) that skips personalisation entirely.

Watch for N+1 on exam, and the JSON district filter – show me the generated SQL for the district case.

Verify: query count must stay flat as notification count grows.

T26 — Feed UI

[CONTEXT PRIMER]

Today I am working on: the feed UI.

Notification feed Livewire component and Blade views.

Card: title (locale-aware), organisation, eligibility badge (green "You are eligible" / amber "3 of 4 match" / grey / none when logged out), vacancies, last date with RED highlight inside 3 days, Details + Save.

Requirements:

- Filter bar: horizontally scrollable chips on mobile, not a dropdown
- "Only show what I am eligible for" toggle
- Infinite scroll via Livewire with a skeleton loader
- Empty state that HELPS: "No notifications match. Try removing the district filter."
- Native ad slot every 5th card (component placeholder)
- Filter state in the query string so a filtered feed is shareable
- 360px first; Telugu must not clip

Show me the card with a long Telugu title so I can check the layout holds.

T27 — Notification detail

[CONTEXT PRIMER]

Today I am working on: notification detail page.

/[locale]/notifications/{slug}:

Sections: title + organisation, eligibility panel (each criterion with a tick or cross AND the reason for each failure), vacancy breakdown table, full eligibility, dates timeline with days-remaining, fee table by category, salary, how to apply, official PDF link, apply link, related notifications.

MANDATORY:

- Visible "Last verified: [date]" linking to the official source
- Disclaimer: always confirm against the official notification before applying
- "Report an error" button filing to a moderation queue
- JobPosting structured data
- Sticky bottom bar on mobile: Apply, Save

Full files.

Done when: the "verify officially" message cannot be missed on mobile.

T28 — Save and reminders

[CONTEXT PRIMER]

Today I am working on: save and deadline reminders.

1. Save/unsave toggle on card and detail, optimistic UI
2. /[locale]/saved grouped by deadline proximity
3. On save, schedule reminders at D-3 and D-1 before apply_end_date
4. SendDeadlineReminders scheduled job – IDEMPOTENT, must never double-send
5. Skip reminders for notifications already marked applied
6. Reminder copy in the user's preferred locale

Full files plus a test proving no duplicates when the job runs twice.

T29-T31 — Historical backfill (*no code*)

Load 12 months of past notifications through Filament. Grind work — and it is what makes the site look alive on day one instead of empty.

M2 test checklist

- ☐ Eligibility coverage above 95%
- ☐ SC + age relaxation case passes

- ☐ PwD + category stacking passes
- ☐ Null date_of_birth returns `unknown`, does not crash
- ☐ `partial` notifications are still visible, never hidden
- ☐ Profile change clears the eligibility cache
- ☐ Feed query count flat as notifications grow
- ☐ Feed filters survive a page refresh (query string)
- ☐ Long Telugu title does not clip the card
- ☐ Reminder job run twice sends once
- ☐ Detail page shows "Last verified" above the fold on mobile

M2 exit gate

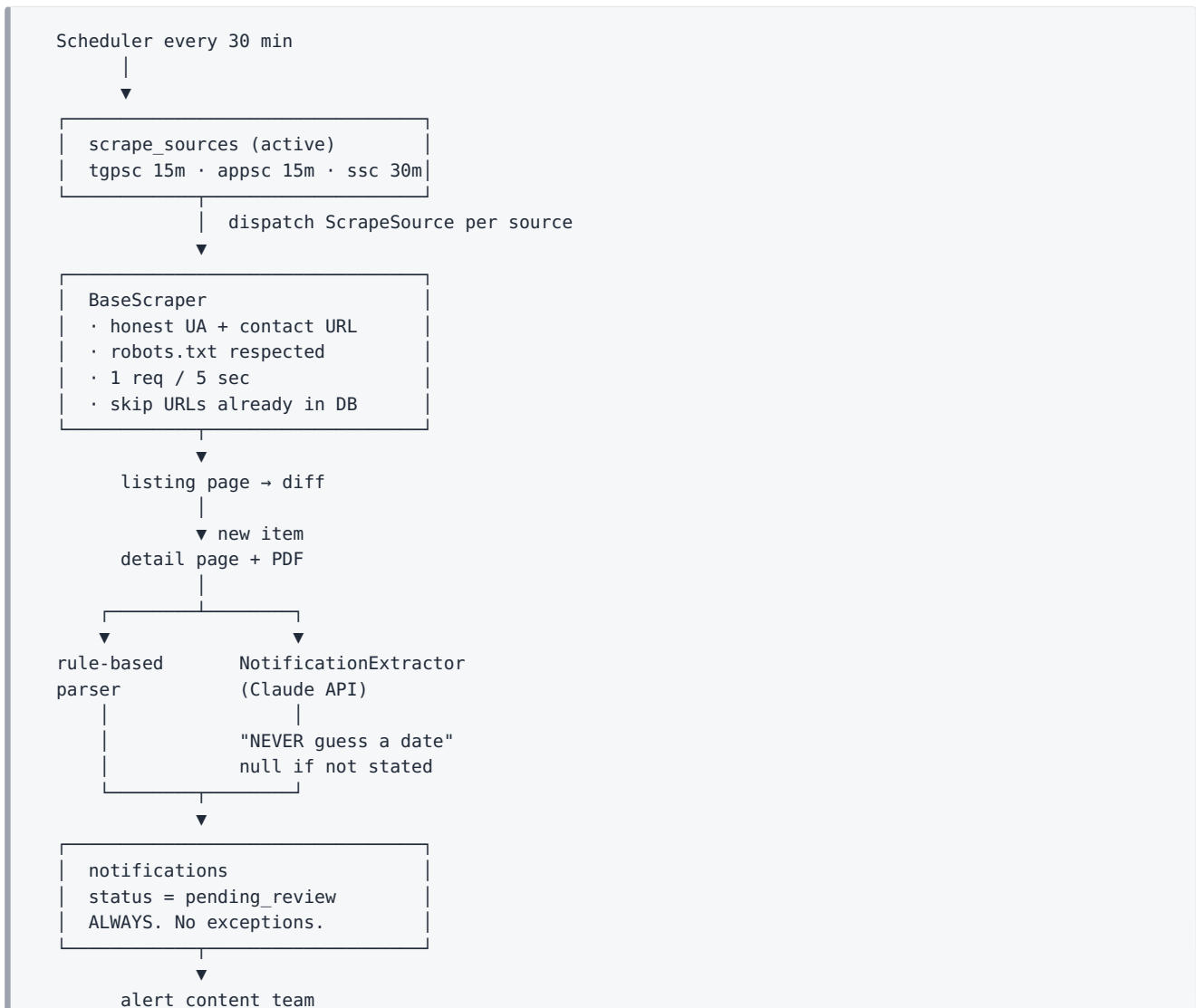
- ☐ Personalised, eligibility-tagged feed working
 - ☐ 12 months of historical notifications loaded
 - ☐ Save and deadline reminders working
 - ☐ JobPosting validates on notification pages
-

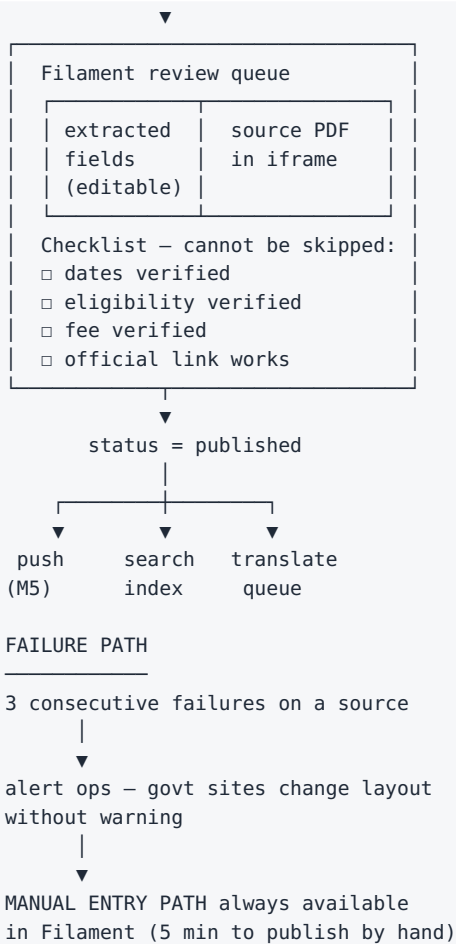
M3 — INGESTION

సారాంశం: స్క్రాపర్లు డ్రాఫ్ట్ మాత్రమే తెస్తాయి. మనిషి తనిఖీ లేకుండా ఏదీ పబ్లిష్ కాదు.

Purpose	Content velocity — publish within 60 minutes of an official release
Tasks	T32–T39 · 6 working days
Depends on	M2
Unlocks	M5 (alerts need fresh notifications)
Exit signal	Scrape → review → publish in under 5 minutes, end to end

M3 data flow





M3 file tree

```

app/Services/Scrapers/BaseScraper.php
app/Services/Scrapers/TgpscScraper.php
app/Services/Scrapers/AppscScraper.php
app/Services/NotificationExtractor.php
app/Jobs/ScrapeSource.php
app/Console/Commands/ScrapeRun.php
app/Filament/Pages/NotificationReviewQueue.php
app/Notifications/ScraperBrokenNotification.php

tests/Feature/Scrapers/TgpscScraperTest.php
tests/fixtures/tgpsc-listing.html
tests/fixtures/tgpsc-detail.html
tests/fixtures/notification-sample.txt

```

M3 build sequence

T32 — TGPSC scraper

[CONTEXT PRIMER]

Today I am working on: the TGPSC notification scraper.

Build an abstract BaseScraper plus TgpscScraper.

BaseScraper handles:

- fetch with 30s timeout and an HONEST user agent (SadhanaBot/1.0 with a contact URL)
- skip URLs already in the database
- create notifications with status = pending_review ALWAYS. Never auto-publish – a wrong date costs a user their year.
- track consecutive_failures on the source
- alert ops after 3 consecutive failures, because government sites change layout without warning

TgpscScraper parses the listing at tgpsc.gov.in and each detail page.
Current HTML structure:

[paste the actual HTML you copied from the site]

Requirements:

- respect robots.txt, rate limit 1 request per 5 seconds
- store FACTS (dates, vacancy counts, eligibility), not verbatim document copies
- always keep source_url so we deep-link to the official page
- fail loudly, never silently

Full files plus a test using a saved HTML fixture.

T33 — APPSC scraper

Same prompt, different site and HTML. Once two scrapers exist, the third onward is a 30-minute job.

T34 — AI extraction

[CONTEXT PRIMER]

Today I am working on: AI extraction from notification PDFs.

App\Services\NotificationExtractor takes PDF text, returns structured data via the Anthropic API.

Fields: title, organisation, total_vacancies, min_qualification, min_age, max_age, age_relaxation (object by category), apply_start_date, apply_end_date, application_fee (object by category), salary_min, salary_max.

Non-negotiable rules baked into the prompt:

- Return ONLY valid JSON, no markdown fences, no preamble
- null for anything not clearly stated
- NEVER infer or guess a date
- Output ALWAYS lands in pending_review for human verification

Also handle: parsing that survives stray markdown, retry on malformed JSON, and a hard failure that LOGS rather than saving garbage.

Full file plus a test with a real notification's text as a fixture.

T35 — Review queue

[CONTEXT PRIMER]

Today I am working on: the notification review queue.

A Filament page for reviewing scraped notifications:

- Split view: extracted fields left, source PDF in an iframe right
- Every field editable inline
- A checklist that MUST be ticked before publish: dates verified, eligibility verified, fee verified, official link works
- Publish / Reject / "Needs more info" actions
- Sorted oldest first
- A visible SLA warning when an item has been pending over 2 hours

Full files.

T36-T39 — Source expansion

Add SSC, RRB, IBPS, NCS scrapers using the T32 pattern. Configure frequencies per Vol 2 Chapter 11.4.

M3 test checklist

- ☐ Scraper skips URLs already in the database
- ☐ Every scraped record lands in `pending_review`, never `published`
- ☐ 3 consecutive failures fires an ops alert
- ☐ Rate limiting respected (check the request log)
- ☐ Extractor returns null rather than guessing a missing date
- ☐ Malformed AI JSON is logged, not saved
- ☐ Review checklist cannot be bypassed
- ☐ Manual entry path publishes in under 5 minutes
- ☐ `source_url` always populated

M3 exit gate

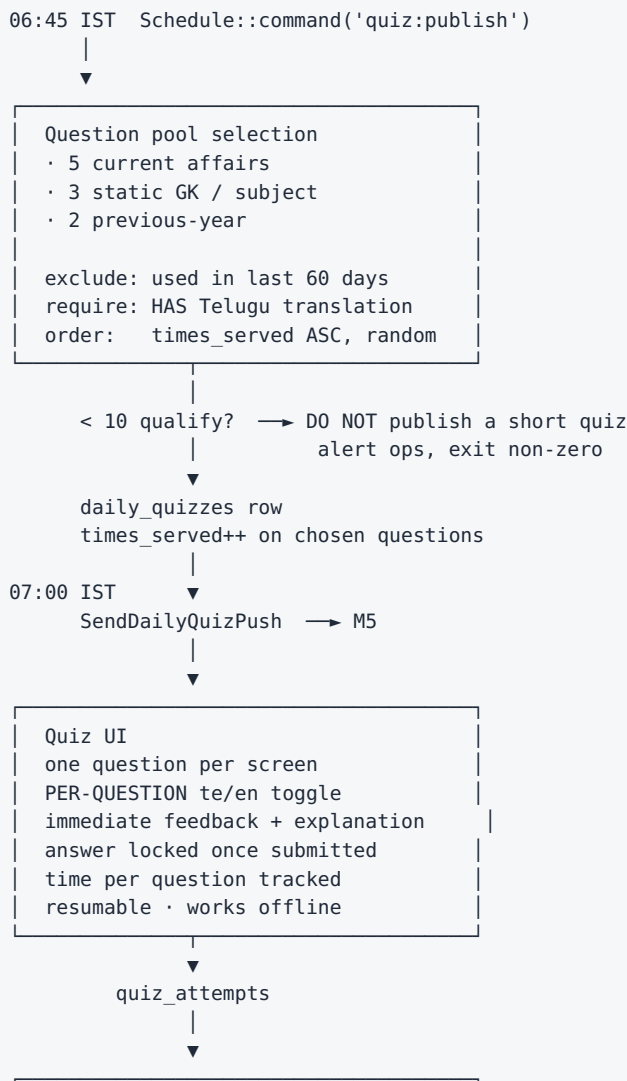
- ☐ Two scrapers running on schedule
- ☐ Review queue in daily use
- ☐ Publish latency under 60 minutes, measured on a real notification
- ☐ Manual fallback tested with a scraper deliberately disabled

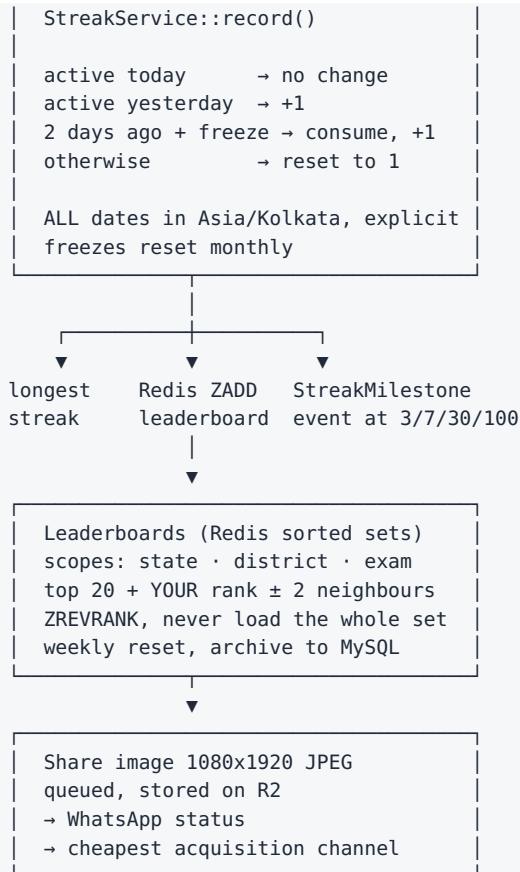
M4 — QUIZ AND STREAKS

సారాంశం: రిటెన్షన్ ఇంజిన్. ఇది పని చేయకపోతే మిగతా అంతా వృధా.

Purpose	The 7 AM habit. Without it there is no daily user and no ad inventory.
Tasks	T40-T50 · 12 working days
Depends on	M0
Unlocks	M5
Exit signal	Quiz publishes automatically at 06:45 and streaks survive midnight, month and timezone boundaries

M4 data flow





M4 file tree

```

app/Console/Commands/PublishDailyQuiz.php
app/Services/StreakService.php
app/Services/LeaderboardService.php
app/Services/QuizResultImage.php
app/Livewire/Quiz/TakeQuiz.php
app/Livewire/Quiz/QuizResult.php
app/Livewire/Quiz/Leaderboard.php
app/Events/StreakMilestoneReached.php
app/Listeners/AwardStreakMilestone.php
app/Filament/Resources/QuestionResource.php
app/Filament/Widgets/QuestionPoolDepth.php
app/Imports/QuestionCsvImport.php
app/Notifications/QuizPoolLowNotification.php

resources/views/quiz/take.blade.php
resources/views/quiz/result.blade.php
resources/views/quiz/partials/question-card.blade.php
resources/views/quiz/partials/language-toggle.blade.php

tests/Unit/StreakServiceTest.php      ← timezone edge cases
tests/Feature/PublishDailyQuizTest.php
tests/Feature/QuizFlowTest.php
  
```

M4 build sequence

T40 — Question bank

[CONTEXT PRIMER]

Today I am working on: the question bank.

Filament resource for Question:

- Bilingual question text, options, explanation – tabs per locale
- Correct answer as a radio over the ENGLISH options, INDEX-LINKED so translations stay aligned
- Fields: exam, subject, topic, difficulty, is_current_affairs, source_year
- Bulk CSV import with a template download and PER-ROW validation errors
- Filters: missing Telugu, never served, subject, difficulty
- A dashboard widget showing pool depth by category – running out of questions breaks the daily quiz

Full files.

Verify: import 50 rows including deliberately broken ones — bad rows report individually without aborting the import.

T41 — Daily quiz generation

[CONTEXT PRIMER]

Today I am working on: daily quiz generation.

quiz:publish artisan command:

- Composition: 5 current affairs, 3 static, 2 previous-year
- Exclude anything used in the last 60 days
- ONLY questions that HAVE a Telugu translation – a half-English quiz breaks the promise of the product
- Rotate the pool: order by times_served ASC, then random
- If fewer than 10 qualify: do NOT publish a short quiz. Alert ops, exit non-zero.
- Idempotent – running twice for the same date is a no-op
- Increment times_served on chosen questions
- A --date option for backfilling

Scheduler at 06:45 IST. Full file plus tests for the pool-depletion path and the idempotency path.

T42 — Quiz UI

[CONTEXT PRIMER]

Today I am working on: the daily quiz UI.

Quiz-taking Livewire component:

- One question per screen, progress bar
- A PER-QUESTION language toggle (te/en) – a user may know the Telugu term but read the English option faster; do not force one language for the whole quiz

- Immediate feedback with explanation after answering
- Answer locked once submitted
- Track time per question
- Resume where they left off if they close mid-quiz
- Works offline once loaded (cache the question set)
- Large tap targets – used one-handed, on a bus

Result screen: score, streak update, weak-area tag, leaderboard position, share button, review answers.

Full files.

T43 — Streak service

[CONTEXT PRIMER]

Today I am working on: streaks.

App\Services\StreakService::record(User).

- Already active today → no change
- Last active yesterday → increment
- Last active 2 days ago AND freezes_left > 0 → consume a freeze and increment. Users have exams, travel, emergencies – a streak that punishes real life gets abandoned and never restarts.
- Otherwise → reset to 1
- Update longest_streak
- Reset freezes to 1 at the start of each month
- Write to a Redis sorted set for the weekly leaderboard
- Fire StreakMilestoneReached at 3, 7, 30, 100

ALL date logic in Asia/Kolkata EXPLICITLY, not server default.

Exhaustive Pest tests: freeze path, month rollover, a user acting at 23:59 then 00:01, timezone correctness, first-ever attempt.

Verify: travel the clock across midnight and month boundaries in tests.

T44 — Leaderboards

[CONTEXT PRIMER]

Today I am working on: leaderboards.

Weekly leaderboards on Redis sorted sets:

- Three scopes: state-wide, district, exam cohort
- Top 20 PLUS the user's own rank with two neighbours above and below – everyone should see themselves, not just the top 20
- Weekly reset, previous week archived to MySQL
- Rank lookup via ZREVRANK, never load the whole set
- Livewire component with a scope switcher

Full files.

Verify: seed 10,000 users — the page must still load under 300ms.

T45 — Share image

[CONTEXT PRIMER]

Today I am working on: the shareable quiz result image.

This is my cheapest acquisition channel – a WhatsApp-status image that makes people ask "what app is that".

A service generating a 1080x1920 JPEG with Intervention Image:

- Score, streak count, date, sadhana.study wordmark
- Telugu rendered with Noto Sans Telugu (bundle the TTF)
- Designed for WhatsApp status: bold, readable at THUMBNAIL size, safe margins
- Generated on a queue, not in the request
- Stored on R2, URL returned
- Web Share API where supported, download fallback

Describe the visual layout before you code it so I can approve the design.

Verify: actually post one to WhatsApp status and look at it on a phone.

T46-T50 — Content sprint

500 questions, both languages. 60 days of quizzes pre-scheduled. This is the largest content push in the project and it is what lets you sleep during launch week.

[CONTEXT PRIMER]

Today I am working on: quiz question drafting.

Generate 20 MCQs for [SUBJECT] at [DIFFICULTY] for Telugu-state government exam aspirants.

Per question: question, 4 options, correct index, 2-sentence explanation. English first, then Telugu.

Rules:

- Match the style of actual TGPSC/APPSC/SSC papers
- Only verifiable facts – no opinion, no disputed answers
- Distractors must be PLAUSIBLE, not obviously wrong
- Exam terms, "Group 2", abbreviations untranslated in Telugu
- FLAG any question where you are less than fully certain of the answer

Output as CSV matching my import template.

Verify: you personally check every answer key. Every one.

M4 test checklist

- ☐ `quiz:publish` twice for the same date is a no-op
- ☐ Empty pool alerts and exits non-zero, publishes nothing

- ☐ Only questions with Telugu translations are selected
- ☐ Question not repeated within 60 days
- ☐ Quiz resumes after closing the browser mid-way
- ☐ Language toggle works per question, not globally
- ☐ Streak: acting at 23:59 then 00:01 increments once
- ☐ Streak: freeze consumed at exactly 2 days gap
- ☐ Streak: freezes reset on the 1st of the month
- ☐ Leaderboard under 300ms at 10,000 users
- ☐ Share image readable at thumbnail size

M4 exit gate

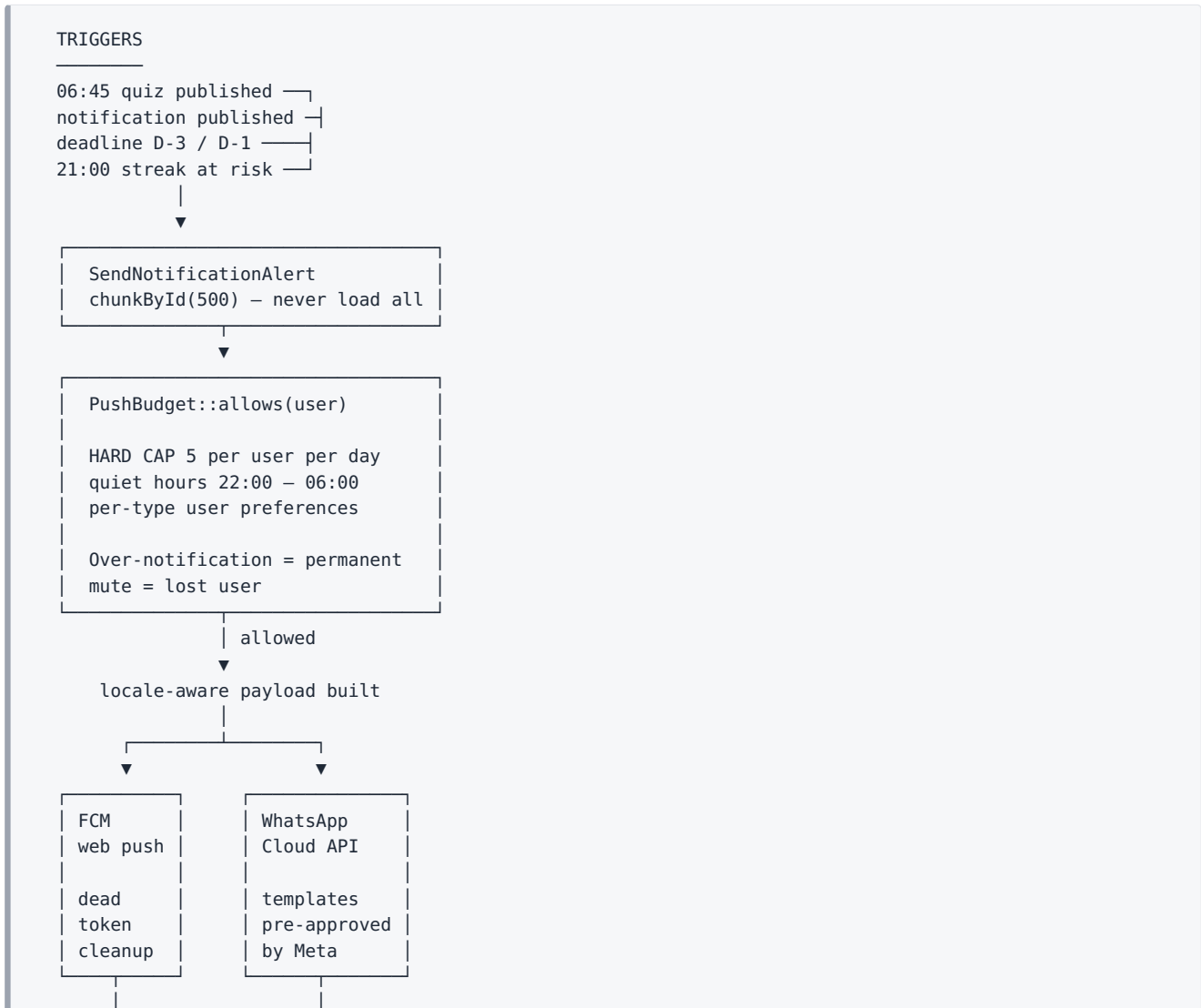
- ☐ Quiz publishes automatically at 06:45
 - ☐ 500 questions in the bank, all answer keys human-verified
 - ☐ 60 days scheduled
 - ☐ Streaks correct across every boundary case
-

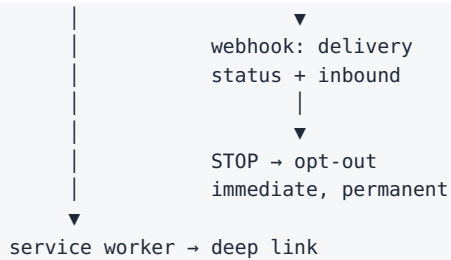
M5 — PUSH AND WHATSAPP

సారాంశం: వాట్సాప్ మన బలమైన మార్గం. అతిగా పంపితే అందరూ ఆఫ్ చేసేస్తారు.

Purpose	Distribution. The 7 AM push is what makes the habit a habit.
Tasks	T51–T54 · 5 working days
Depends on	M2, M4
Unlocks	M6, M7
Exit signal	Push and WhatsApp both delivering, daily budget enforced server-side

M5 data flow





DISPATCH SAFETY

50,000 users at once will melt the app tier on notification day.

↓
STAGGERED WAVES, queued on 'push' priority
idempotent – safe to re-run

M5 file tree

```

app/Services/PushBudget.php
app/Services/WhatsAppService.php
app/Jobs/SendPush.php
app/Jobs/SendNotificationAlert.php
app/Jobs/SendDailyQuizPush.php
app/Jobs/SendStreakReminders.php
app/Http/Controllers/WhatsAppWebhookController.php
app/Livewire/Settings/NotificationPreferences.php
app/Services/Otp/WhatsAppDriver.php      ← completes the M0 stub

public/firebase-messaging-sw.js
resources/views/settings/notifications.blade.php

docs/whatsapp-templates.md              ← exact text for Meta approval
  
```

M5 build sequence

T51 — Web push

[CONTEXT PRIMER]

Today I am working on: web push.

FCM web push for a Laravel PWA:

1. Permission request at the RIGHT moment – after the first completed quiz, never on first page load
2. Token storage, deactivation on failure
3. SendPush queued job with retry and dead-token cleanup
4. A PushBudget service enforcing a HARD CAP of 5 pushes per user per day. Over-notification is how you get permanently muted, and a muted user is a lost user.
5. Per-type preferences: daily quiz, new notifications, deadlines, streak warnings
6. Locale-aware payloads

7. Service worker handler with deep links

Full files including the complete service worker.

Verify: send yourself 6 pushes in a day — the 6th must be suppressed **server-side**.

T52 — WhatsApp

[CONTEXT PRIMER]

Today I am working on: WhatsApp Cloud API.

I have Meta Cloud API experience from another project. For Sadhana:

1. WhatsAppService: send template messages, send session messages, verify webhooks
2. Message templates for Meta approval – draft the EXACT text in English and Telugu for: daily quiz link, new notification alert, deadline reminder, OTP
3. Webhook handling for delivery status and inbound messages
4. Opt-in and opt-out, with opt-out honoured IMMEDIATELY and PERMANENTLY
5. Rate limiting to Meta's tier limits with backoff

Full files plus the exact template submission text.

Verify: reply STOP — opt-out must stick permanently.

T53 — Dispatch at scale

[CONTEXT PRIMER]

Today I am working on: notification dispatch at scale.

SendNotificationAlert:

- Target users whose exam preferences match
- chunkById(500), NEVER load all users into memory
- Respect PushBudget, per-type preferences, quiet hours 22:00-06:00
- Locale-aware content per user
- Dispatch to the 'push' priority queue
- STAGGERED WAVES – 50,000 at once will melt the app tier on notification day
- Idempotent: safe to re-run

Full file plus a test with 10,000 seeded users asserting memory stays flat.

T54 — Streak reminders

[CONTEXT PRIMER]

Today I am working on: streak reminder push at 21:00 IST.

Only for users with an active streak who have NOT been active today.

Copy tone matters enormously: my users are already under exam stress and

family pressure. Encouraging, never guilt-inducing. No "you're about to lose everything".

Draft 5 variants in English and Telugu, tell me which one you would ship and why. Then build the scheduled job.

Verify: read the copy aloud. If it would make a stressed person feel worse, reject it.

M5 test checklist

- ☐ 6th push in a day suppressed server-side
- ☐ Quiet hours respected
- ☐ Per-type preferences honoured
- ☐ Dead FCM tokens deactivated automatically
- ☐ WhatsApp STOP opts out permanently
- ☐ Webhook signature verified
- ☐ Dispatch memory flat at 10,000 users
- ☐ Re-running dispatch sends nothing twice
- ☐ Payload locale matches the user's preference

M5 exit gate

- ☐ Push and WhatsApp both delivering
 - ☐ Daily budget enforced server-side
 - ☐ Meta templates approved
 - ☐ Streak reminder copy reviewed for tone
-

M6 — PWA AND PERFORMANCE

సారాంశం: చౌక ఫోన్‌లో పని చేయకపోతే మార్కెట్ లేదు.

Purpose	Reach on a ₹8,000 phone. Survive notification day.
Tasks	T55–T57 · 5 working days
Depends on	M1, M2, M4
Unlocks	M7
Exit signal	Every budget in Vol 2 Chapter 1.3 met on a throttled profile, 50x load test survived

M6 data flow

CACHING STRATEGY (service worker)

```

app shell           → cache on install
exam pages          → stale-while-revalidate
feed                → network-first, cache fallback
fonts, static assets → cache-first
quiz question set   → cached so a quiz survives a dropped connection
everything else      → offline fallback page IN THE USER'S LANGUAGE
  
```

PERFORMANCE BUDGETS

```

LCP   < 2.5s   on 3G, Moto G4 profile
TTI    < 3.5s
JS     < 100 KB gzipped  ← enforced in CI, build fails above
server < 500ms p95 dynamic
       < 200ms p95 cached
  
```

NOTIFICATION-DAY DEGRADATION LADDER

```

p95 > 2s
|
▼ shed 1
personalised feed OFF → serve anonymous cached feed to everyone
|
▼ shed 2
eligibility badges OFF → notifications still visible
|
▼ shed 3
static fallback page: essential facts only, no personalisation

Degraded is ALWAYS better than absent.
  
```

M6 file tree

```
public/manifest.json
public/sw.js
app/Livewire/InstallPrompt.php
resources/views/offline.blade.php
app/Http/Middleware/DegradedMode.php
config/degradation.php

tests/k6/baseline.js
tests/k6/spike.js
docs/notification-day-runbook.md
scripts/check-bundle-size.js      ← from M0 CI
```

M6 build sequence

T55 — PWA

```
[CONTEXT PRIMER]
Today I am working on: PWA.

Build:
1. manifest.json – name, Telugu short_name, icons, standalone, theme
   colour, start_url with locale
2. A service worker:
   - app shell cached on install
   - stale-while-revalidate for exam pages
   - network-first with cache fallback for the feed
   - cache-first for fonts and static assets
   - offline fallback page IN THE USER'S LANGUAGE
   - cached quiz question sets so a quiz survives a dropped connection
3. Install prompt after the THIRD visit, never the first
4. Update handling: notify, do NOT force-reload mid-quiz

Full files including the complete service worker.
```

Verify: install, airplane mode, open saved content — must be useful, not a browser error.

T56 — Performance pass

```
[CONTEXT PRIMER]
Today I am working on: performance.

My Lighthouse results on a throttled 3G Moto G4 profile:

[paste the full report]

Budgets: LCP < 2.5s, TTI < 3.5s, JS < 100 KB gzipped, server p95 < 500ms.

Give me a PRIORITISED fix list with the actual code changes, ranked by
impact-per-hour-of-work. Ignore anything that will not move a real user
on a real cheap phone.
```

T57 — Load test and degradation runbook

[CONTEXT PRIMER]

Today I am working on: load testing.

k6 scripts simulating notification-day traffic:

- Baseline: 500 concurrent
- Spike: 25,000 concurrent over 3 minutes (my traffic goes 20-50x when a major notification drops)
- Mixed journey: 60% feed, 25% exam pages, 10% quiz, 5% search
- Logged-in and anonymous mix

Also the DEGRADATION STRATEGY: when p95 exceeds 2s, what do I shed first, second, third? I would rather serve a degraded site than no site.

Scripts plus a written runbook I can follow at 3 AM.

M6 test checklist

- ☐ Installable on Android Chrome
- ☐ Offline: saved material and last-seen feed open
- ☐ Offline page renders in the user's language
- ☐ Quiz survives a dropped connection mid-way
- ☐ Update does not force-reload mid-quiz
- ☐ LCP under 2.5s throttled
- ☐ JS under 100 KB gzipped — CI enforces it
- ☐ 50x spike survived
- ☐ Each degradation level tested manually

M6 exit gate

- ☐ All performance budgets met
- ☐ Load test passed at 50x
- ☐ Degradation runbook written and rehearsed

M7 — COMPLIANCE AND LAUNCH

సారాంశం: చట్టపరమైన అంశాలు పూర్తి చేయకుండా లైవ్ కి వెళ్లకూడదు.

Purpose	Legal survival, then launch
Tasks	T58-T62 · 8 working days
Depends on	Everything
Unlocks	Revenue
Exit signal	Live, monitored, and a stranger has completed a quiz

M7 data flow

DPDP CONSENT

signup



itemised checkboxes, in the user's language,
purpose stated PER ITEM



store: consent version + IP + timestamp

RIGHTS

"Download my data"



JSON export of
everything tied
to the user

"Delete my account"



confirmation + grace period



hard-delete personal data
within 30 days



retain ONLY anonymised
aggregates

RETENTION

inactive 3 years

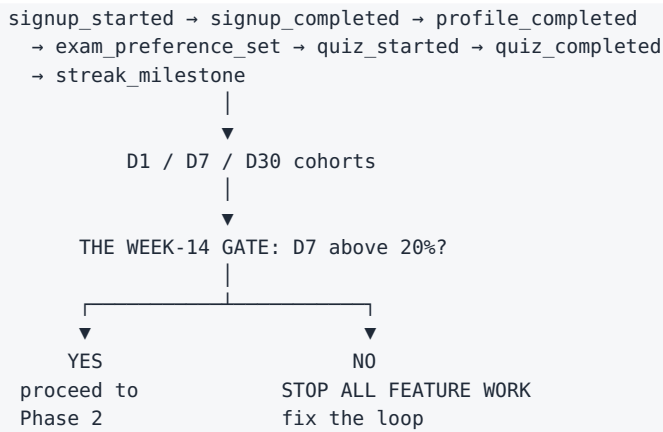


anonymised automatically

AUDIT

every admin READ of user
data logged

ANALYTICS — no PII in any payload



M7 file tree

```

app/Services/DataExportService.php
app/Services/AccountDeletionService.php
app/Console/Commands/AnonymiseInactive.php
app/Http/Middleware/AuditAdminDataAccess.php
app/Livewire/Settings/PrivacyControls.php
app/Services/Analytics/EventTracker.php
app/Filament/Pages/AnalyticsDashboard.php

resources/views/legal/privacy.blade.php    ← te + en
resources/views/legal/terms.blade.php
resources/views/legal/copyright.blade.php
resources/views/legal/takedown.blade.php

docs/launch-checklist.md
  
```

M7 build sequence

T58 — DPDP compliance

[CONTEXT PRIMER]

Today I am working on: DPDP Act 2023 compliance.

I store name, phone, date of birth, category, qualification and district for lakhs of Indian users. Build:

1. Consent management: itemised checkboxes at signup in the user's language, purpose stated per item; store consent version, IP, timestamp
2. A WORKING "Download my data" producing a JSON export of everything tied to the user
3. A WORKING "Delete my account": hard-delete personal data within 30 days, retain only anonymised aggregates, with a confirmation flow and a grace period
4. A retention job anonymising accounts inactive for 3 years
5. An audit log on every admin READ of user data
6. Privacy policy and terms in English and Telugu, written plainly enough for a 22-year-old to actually understand

Full files. FLAG anything where I need an actual lawyer rather than code.

Verify: create a test account, export, delete, confirm the data is gone from the database.

T59 — Legal pages (*no code*)

Privacy policy, terms, copyright policy, takedown process — both languages, real contact address, stated 48-hour SLA.

Done when: all four live and linked from the footer on every page.

T60 — Analytics

[CONTEXT PRIMER]

Today I am working on: analytics.

Event tracking for the metrics that actually matter (Vol 1 Chapter 10).

Events: signup_started, signup_completed, profile_completed, exam_preference_set, quiz_started, quiz_completed, streak_milestone, notification_viewed, notification_saved, eligibility_checked, search_performed, share_clicked, push_permission_granted

Requirements:

- NO personally identifiable data in any event payload – DPDP
- Cohort retention: D1, D7, D30
- A funnel view: organic visit → signup → profile → first quiz
- A simple internal dashboard, not a paid third-party tool

Full files.

T61 — Beta (*no code*)

20 real users, recruited in person at coaching centres. Watch them use it. **Do not explain anything** — every time you need to explain, that is a bug. Ship one fix per day based on what you saw.

Done when: a new user reaches their first completed quiz without you saying a word.

T62 — Launch

Work the checklist in Vol 2 Chapter 19. Launch on a day a major notification is expected — ride the spike rather than fighting for attention on a quiet Tuesday.

M7 test checklist

- ☐ Consent stored with version, IP, timestamp

- ☐ Data export returns everything tied to the user
- ☐ Account deletion actually removes rows, verified in the database
- ☐ Anonymisation job runs on inactive accounts
- ☐ Admin data reads appear in the audit log
- ☐ No PII in any analytics payload
- ☐ D1/D7/D30 visible on the dashboard
- ☐ All four legal pages live in both languages

M7 exit gate

- ☐ DPDP: consent, export, deletion all working
 - ☐ Legal pages published
 - ☐ Analytics showing real retention numbers
 - ☐ 20 beta users onboarded, their friction fixed
 - ☐ Launch checklist complete
-

Reusable prompt library

సారాంశం: ఈ ప్రాంప్ట్లు ప్రతి వారం వాడతావు.

Pre-merge review

Review this before I merge.

[code]

Check, in this order:

1. N+1 queries
2. Cache keys missing `app()->getLocale()`
3. Missing authorization
4. Unvalidated user input
5. Anything that breaks on 3G or a 360px screen
6. Anything that breaks when the Telugu string is 30% longer

Rank by severity. Skip style nitpicks – Pint handles those.

Debugging

Broken.

EXPECTED: [x]

GOT: [y]

ERROR:

[full stack trace]

CODE:

[the file]

TRIED: [a, b]

Find the cause. Show me the fix, not a rewrite.

Tests for existing code

Write Pest tests for this:

[code]

Cover: happy path, every edge case you can find, failure modes, and specifically the Telugu/English locale variants.

I want to know what breaks, not to feel good about coverage numbers.

Performance investigation

Slow: [endpoint], [n]ms p95.

CODE: [controller, service, query]

DEBUGBAR: [query log]

Rank the causes by impact. Fix for the top one first.

Telugu content review

Review this Telugu translation for a government exam portal:

EN: [english]

TE: [telugu]

Check:

- Does it sound like everyday Telugu an aspirant uses, or like a machine?
- Are exam terms, "Group 2", and abbreviations left untranslated?
- Is the register right – respectful but not formal-literary?
- Any word here that people would NOT type into Google?

Corrected version if needed.

Module status check

[CONTEXT PRIMER]

Module [Mx]. Status:

DONE: [tasks]

IN PROGRESS: [task, % complete]

BLOCKED: [what and why]

My module exit gate is: [paste it]

Tell me:

1. Am I done – yes or no
2. What is genuinely missing
3. The one thing to do first tomorrow

Be blunt. I would rather cut scope than ship a half-module.

The five decisions you will be tempted to reverse

#	Decision	The temptation	Why you hold
1	Community opens only after 5,000 DAU	"Let's launch with it, it looks better"	An empty forum signals a dead product and permanently damages first impressions
2	No pirated PDFs, ever	"Everyone does it, it would 10x traffic"	One publisher notice ends the business. Traffic you cannot keep is not traffic.
3	Two languages at launch, not eight	"More languages, more users"	Thin pages destroy the SEO engine the whole model depends on
4	Human review on every date and eligibility	"The AI extraction is good enough now"	One wrong date costs a user a year and costs you their trust forever
5	Stop feature work if D7 is under 20% at Week 14	"One more feature will fix retention"	If the habit loop does not work, nothing built on top of it will

End of Volume 4.

Vol 1: why. Vol 2: how. Vol 3: when. Vol 4: this module, start to finish. Open M0 and start with T01.